Honours Linear Algebra - Week 3

21 and 23 September 2026

21 Sept. / Dimension of a sum

Thm.1.31. For finite-dimensional subspaces U, W,

$$\dim(U+W) = \dim U + \dim W - \dim(U \cap W).$$

Proof: take a basis B of $U \cap W$; extend to (B, C) for U, (B, D) for W. Then (B, C, D) spans $U + W$.

$$\sum a_i b_i + \sum c_j u_j + \sum d_k w_k = 0 \Rightarrow \sum d_k w_k \in U \cap W \Rightarrow d_k = 0 \Rightarrow c_j = a_i = 0.$$

Thus (B, C, D) is a basis; count vectors. If $U \cap W = \{0\}$, dimensions add.

Ex.1.52. Distinct planes through 0 in $\mathbb{R}^3$:

$$U + W = \mathbb{R}^3, \qquad \dim(U \cap W) = 2 + 2 - 3 = 1.$$

In $\mathbb{R}^4$, two planes can have zero intersection: $\mathrm{Span}(e_1, e_2)$ and $\mathrm{Span}(e_3, e_4)$.

Def.1.34–35 / several summands.

$$V = \bigoplus_{i=1}^{r} U_i \iff V = \sum_i U_i, \qquad \left[\sum_i u_i = 0,\ u_i \in U_i \Rightarrow u_i = 0\ \forall i\right].$$

Equivalently $U_j \cap \sum_{i \neq j} U_i = \{0\}$ for every j; each v has a unique decomposition $v = \sum_i u_i$.

Pairwise zero intersections are insufficient: the x-axis, y-axis and $y = x$ in $\mathbb{R}^2$ satisfy

$$U_i \cap U_j = \{0\}\ (i \neq j), \qquad e_1 + e_2 - (e_1 + e_2) = 0.$$

Linear maps

Def.2.1–2. $T : V \to W$ over the same field K is linear iff

$$T(au + bv) = aT(u) + bT(v) \quad (a, b \in K).$$

In particular $T(0) = 0$. Notation: $\mathcal{L}(V, W) = \mathrm{Hom}(V, W)$; $\mathrm{End}(V) = \mathcal{L}(V, V)$.

$$(S + T)(v) = S(v) + T(v), \qquad (cT)(v) = cT(v).$$

$\mathcal{L}(V, W)$ is a vector space. Composition is linear; $\mathrm{End}(V)$ is a ring under composition.

Basis images; kernel and image

Prop.2.3–4. A linear map is determined by its values on a basis:

$$v = \sum_j c_j b_j \quad \Rightarrow \quad T(v) = \sum_j c_j T(b_j).$$

Any chosen images of the b_j define a unique linear map. For standard bases,

$$T : K^n \to K^m, \qquad A = \begin{pmatrix} T(e_1) & \cdots & T(e_n) \end{pmatrix} \in M_{m\times n}(K), \quad T(x) = Ax.$$

Images go in the **columns**.

Examples. Zero, identity, inclusion $U \hookrightarrow V$, restriction $T|_U$; also

$$\begin{aligned} \mathrm{ev}_{t_0} : K_n[t] &\to K, & p &\mapsto p(t_0), \\ \mathrm{Tr} : M_n(K) &\to K, & A &\mapsto \sum_i a_{ii}, \\ D : C^\infty(\mathbb{R}) &\to C^\infty(\mathbb{R}), & f &\mapsto f'. \end{aligned}$$

Fourier coefficients are linear too: $\widehat{af + bg}(k) = a\hat{f}(k) + b\hat{g}(k)$.

Def.2.5–6.

$$\ker T = \{v : T(v) = 0\} \le V, \qquad \mathrm{Im}\, T = T(V) = \{T(v) : v \in V\} \le W.$$

Closure follows from linearity. For D above, $\ker D = \{\text{constant functions}\}$.

23 Sept. / Injective, surjective

Def.2.8–10.

$$\begin{aligned} T \text{ injective} &\iff T(u) = T(v) \Rightarrow u = v \iff \ker T = \{0\}, \\ T \text{ surjective} &\iff \mathrm{Im}\, T = W, \\ T \text{ isomorphism} &\iff T \text{ injective and surjective.} \end{aligned}$$

Kernel test: $T(u) = T(v) \Rightarrow u - v \in \ker T$.

Injective $\iff$ preserves every independent list. Key step:

$$\sum_i a_i T(v_i) = 0 \Rightarrow T\left(\sum_i a_i v_i\right) = 0 \Rightarrow \sum_i a_i v_i = 0 \Rightarrow a_i = 0.$$

Reverse: apply preservation to singletons $\{v\}$, $v \neq 0$. Surjective $\iff$ images of a spanning set span W.

Ex.2.9–11. Inclusion: injective. $T : V \to T(V)$: onto. If $V = U \oplus W$,

$$\pi_W(u + w) = w, \qquad \ker \pi_W = U, \quad \mathrm{Im}\, \pi_W = W.$$

Isomorphisms; matrix representation

Prop.2.11–13. An isomorphism has a unique linear inverse:

$$T^{-1}T = I_V, \qquad TT^{-1} = I_W, \qquad (S \circ T)^{-1} = T^{-1} \circ S^{-1}.$$

Linearity: write $w_i = T(v_i)$, then $T^{-1}(aw_1 + bw_2) = av_1 + bv_2$. For a basis $\mathcal{B} = (b_1, \ldots, b_n)$,

$$\psi_{\mathcal{B}} : V \xrightarrow{\sim} K^n, \quad v \mapsto [v]_{\mathcal{B}}, \qquad \psi_{\mathcal{B}}^{-1}(c) = \sum_i c_i b_i.$$

Thus finite-dimensional $V \cong W \iff \dim_K V = \dim_K W$.

Ex.2.12–16.

$$K_n[t] \cong K^{n+1}, \quad M_{m\times n}(K) \cong K^{mn}, \quad \mathbb{C} \cong \mathbb{R}^2 \text{ over } \mathbb{R} \ (a + bi \mapsto (a, b)).$$

The xy-plane in $\mathbb{R}^3$ is isomorphic to $\mathbb{R}^2$, though not equal to it. $U \times W = (U \times \{0\}) \oplus (\{0\} \times W)$.

Def.2.14. Let $\mathcal{B}$ be a basis of V, $\mathcal{C}$ a basis of W. Following the slides, source basis upstairs:

$$M = [T]_{\mathcal{C}}^{\mathcal{B}} = ([T(b_1)]_{\mathcal{C}} \ \cdots \ [T(b_n)]_{\mathcal{C}}), \qquad [T(v)]_{\mathcal{C}} = M[v]_{\mathcal{B}}.$$

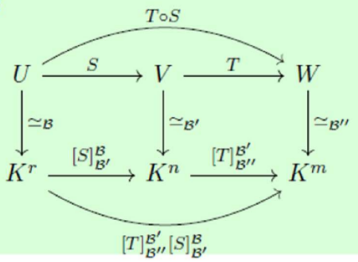
$M = \psi_{\mathcal{C}} \circ T \circ \psi_{\mathcal{B}}^{-1}$: change to coordinates, apply M, change back.

Prop.2.18–19. For $U \xrightarrow{S} V \xrightarrow{T} W$,

$$[T \circ S]_{\mathcal{D}}^{\mathcal{B}} = [T]_{\mathcal{D}}^{\mathcal{C}} [S]_{\mathcal{C}}^{\mathcal{B}}.$$

Rightmost acts first. Standard bases: AB represents $T \circ S$; its j-th column is $T(S(e_j)) = Ab_j$.

Composition of linear maps:



23 Sept., p.12.

Prop.2.16 / trace. Square A, B; invertible Q:

$$\mathrm{Tr}(AB) = \sum_{i,j} a_{ij} b_{ji} = \mathrm{Tr}(BA), \qquad \mathrm{Tr}(Q^{-1}AQ) = \mathrm{Tr}(A).$$

Row operations; one worked matrix

For $T(x) = Ax$, $\ker T = \mathrm{Nul}(A)$, $\mathrm{Im}\, T = \mathrm{Col}(A)$.

Prop.2.22–24. Row operations $R = EA$, E invertible:

$$Rx = 0 \iff Ax = 0, \qquad \sum_j c_j a_j = 0 \iff \sum_j c_j r_j = 0.$$

Same null space and same coefficients in column relations. Column operations AQ with Q invertible preserve the column space.

Ex.2.17–19.

$$A = \begin{pmatrix} 2 & 4 & 0 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 8 & 10 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \sim \underbrace{\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}}_{R}.$$

$Ax = 0$: $x_1 + 2x_2 + x_4 = 0$, $x_3 + x_4 = 0$. Hence

$$x = s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ -1 \\ 1 \end{pmatrix}.$$

These two vectors form a null basis. Pivots in columns $1, 3$:

$$\mathrm{Col}(A) = \mathrm{Span}\left\{ \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 8 \end{pmatrix} \right\}.$$

Use columns of A, not R. Here $a_2 = 2a_1$, $a_4 = a_1 + a_3$; the same relations hold among columns of R.

Cor.2.25 / unique RREF. Pivot columns occur exactly where

$$a_j \notin \mathrm{Span}(a_1, \ldots, a_{j-1}).$$

So their positions are fixed by A. In RREF they are standard unit vectors; all other columns are determined by their unchanged dependence coefficients. Hence two RREFs coincide.

Thm.2.29 / row rank = column rank.

$$\dim \mathrm{Row}(A) = \#\{\text{nonzero echelon rows}\} = \#\{\text{pivots}\} = \dim \mathrm{Col}(A).$$

For this A: rank 2, nullity 2; $T : \mathbb{R}^4 \to \mathbb{R}^3$ is neither injective nor onto.

Rank–nullity

Def.2.28. $\operatorname{rank} T = \dim \operatorname{Im} T$, $\operatorname{nullity} T = \dim \ker T$.

Thm.2.30. $\dim V < \infty$, $T : V \to W$ linear:

$$\boxed{\dim V = \dim \ker T + \dim \operatorname{Im} T.}$$

Proof / lift a basis of the image. Let $(w_1, \dots, w_r)$ be a basis of $\operatorname{Im} T$; choose u_i with $T(u_i) = w_i$, and set $H = \operatorname{Span}(u_1, \dots, u_r)$.

$$\sum_i c_i u_i = 0 \Rightarrow \sum_i c_i w_i = 0 \Rightarrow c_i = 0, \qquad \dim H = r.$$

Likewise $h \in H \cap \ker T \Rightarrow h = \sum c_i u_i$, $\sum c_i w_i = 0 \Rightarrow h = 0$. For any $v \in V$, write $T(v) = \sum b_i w_i$. Then

$$v = \underbrace{\left(v - \sum_i b_i u_i\right)}_{\in \ker T} + \underbrace{\sum_i b_i u_i}_{\in H}.$$

Thus $V = \ker T \oplus H$, $T|_H : H \xrightarrow{\sim} \operatorname{Im} T$; count dimensions.

Consequences. $A \in M_{m\times n}(K)$, rank r:

$$\begin{aligned} \operatorname{nullity} A &= n - r, \\ T \text{ injective} &\iff r = n \quad \text{(all columns pivot)}, \\ T \text{ surjective} &\iff r = m \quad \text{(every row has a pivot)}. \end{aligned}$$

If $\dim V = \dim W < \infty$: injective $\iff$ surjective $\iff$ isomorphism. For injective T, every subspace U satisfies $\dim T(U) = \dim U$ (its basis maps to a basis). Conversely, this equality for all U forces injectivity: use $U = \operatorname{Span}(v)$ for $v \neq 0$.

Ex. / finite-dimensional condition. On $\mathbb{R}[t]$,

$$D(p) = p' \text{ is onto but not injective}, \qquad M(p) = tp \text{ is injective but not onto.}$$

For $Ax = b$: no solution if $b \notin \operatorname{Col}(A)$; otherwise

$$\{x : Ax = b\} = x_p + \operatorname{Nul}(A), \qquad n - r \text{ free parameters.}$$